



NVIDIA NVL72 AI Factory

NVIDIA Enterprise Reference Architecture with NVIDIA GB300 NVL72 and NVIDIA Spectrum-X Networking Platform

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Abstract

The NVIDIA Enterprise Reference Architecture (Enterprise RA) with NVIDIA GB300 NVL72 and NVIDIA Spectrum-X Networking powers enterprise data centers for AI training and inference at massive scale, enabling real-time applications and trillion-parameter models.

This Enterprise RA is designed to solve the most challenging computational problems of today. Certified configurations based on this architecture can be deployed in enterprise data centers around the world.

This Enterprise RA uses the following NVIDIA technologies:

- > [NVIDIA GB300 NVL7272](#)
- > [NVIDIA BlueField®-3 DPUs](#)
- > [NVIDIA ConnectX®-8 SuperNICs](#)
- > [NVIDIA Spectrum™-X Ethernet Platforms](#)
- > [NVIDIA AI Enterprise Software](#)
- > [NVIDIA Mission Control](#)
- > [NVIDIA NVLink and NVLink Switch](#)

This document discusses the components that define the scalable and modular architecture of the Enterprise RA, which is built upon the building blocks of scalable units (SU), each containing 18 compute nodes with 72 GPUs. This Enterprise RA includes details regarding the SU design and specifics of Ethernet fabric topologies, NVLink networking, and storage system specifications.

The Enterprise RA is managed by NVIDIA solutions such as NVIDIA Mission Control, NVIDIA AI Enterprise, NVIDIA® CUDA®, and NVIDIA Magnum IO™. These technologies help keep the system running at the highest levels of availability and performance.

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Overview

The NVIDIA Enterprise RA using 2-4-5-800 (dual plane) node architecture with NVIDIA GB300 NVL72 and NVIDIA Spectrum-X Networking offers a fully integrated, rack-scale solution optimized for the most demanding AI workloads. It documents a modular architecture based on NVIDIA-Certified NVL72 GB300 systems, each with 72 NVIDIA Blackwell Ultra GPUs and 36 NVIDIA® Grace CPUs.

With a liquid-cooled design, each Scalable Unit (a single NVL72 rack) integrates 18 compute trays connected via the fifth generation of NVLink. While each tray (single server or node) can still operate independently as needed, the NVLink interconnect enables GPUs to be dynamically combined, allowing the system to function as a single multi-GPU unit of compute for larger workloads. The design minimizes system bottlenecks through tightly coupled configuration to provide the best performance and application scalability.

A fully tested system scales up to 8 SUs (Scalable Units). Larger clusters can be built based on customer requirements.

The rack solution is delivered as a pre-configured system available through the fulfillment of OEM along with Hardware support. Software support from NVIDIA is based on a per GPU paid subscription of NVIDIA AI Enterprise. While this Enterprise RA document may make the solution look simple, it is highly recommended to utilize NVIDIA Mission Control for operation of the NVIDIA GB300 NVL72.

The NVIDIA GB300 NVL72 and NVIDIA Spectrum-X Platform RA have been developed to support enterprise-grade AI deployments, with the following use cases:

- > **Inference** on today's largest AI models, in real-time
- > **Training** and finetuning of trillion-parameter language models
- > **Data analytics** that's ultra-fast, at massive scale
- > **High-performance computing** for single and mixed precision

For all use cases, this architecture is ideal for multi-user, single tenant workloads. Specifically, the logical design and software is streamlined for deployment and maintenance ease by tailoring the configuration to one where users are all part of the same enterprise, and accounting and access control can be consolidated.

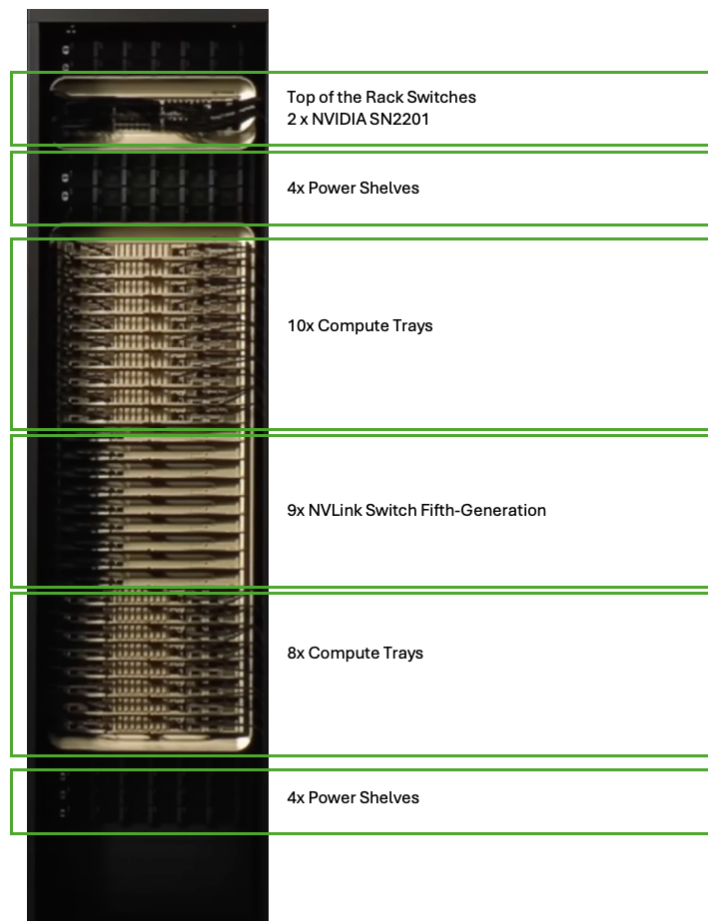
This RA is architected to support deployment of Kubernetes, Slurm, and associated applications and tools for non-virtualized workloads.

System Hardware & Components

This section describes the hardware elements of the Enterprise RA.

NVIDIA GB300 NVL72 The NVIDIA GB300 NVL72 (Figure 1) delivers breakthrough AI performance and can deliver up to a 50x overall increase in AI factory output performance compared to NVIDIA Hopper-based platforms. The NVIDIA Blackwell Ultra GPU architecture provides the latest technologies that bring months of computational effort down to days and hours on some of the largest AI/ML workloads.

Figure 1 NVIDIA GB300 NVL72



To accommodate the massive compute power within a limited space, the GB300 NVL72 rack is liquid cooled, based on the MGX architecture. Some key highlights of the GB300 NVL72 compute rack include:

- > 9 NVSwitch trays for full non-blocking P2P connectivity of all 72 Blackwell Ultra GPUs, with a total aggregated bandwidth of 130 TB/s
- > 2 SN2201 OOB switches for integrated management access of rack components. The GB300 NVL72 in-rack switches use DC power since they are connected to the DC busbar; all other SN2201 switches use AC power
- > Integrated tray-level and rack-level liquid leakage detection
- > 8 power shelves of 33 kW, with each shelf having six 5.5 kW PSUs
- > Full rack requiring up to 142 kW

NVIDIA GB300 NVL Compute Tray

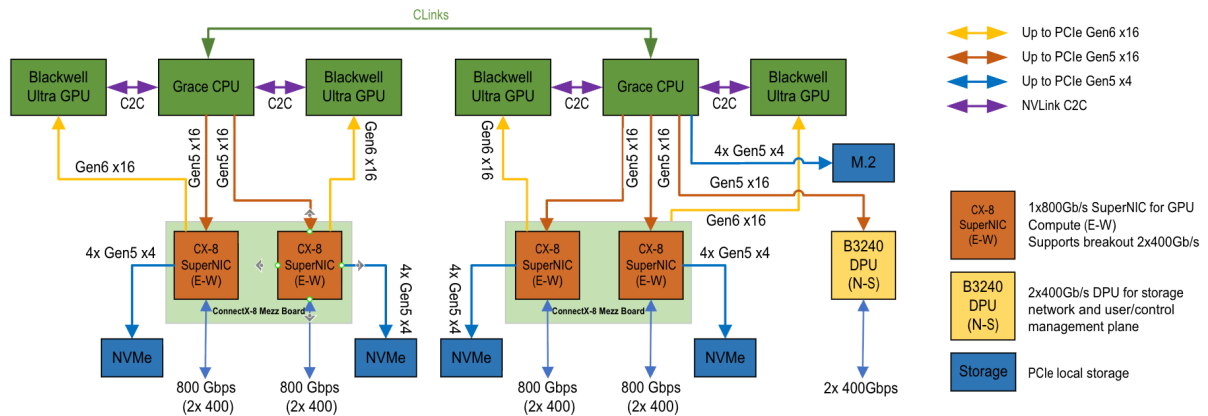
The compute building block for the NVIDIA GB300 NVL72 is the GB300 NVL compute tray/node (Figure 2), with 4 Blackwell Ultra GPUs and 2 Grace CPUs per tray. A logical diagram is shown in Figure 3.

Figure 2 NVIDIA GB300 NVL Compute Tray



The compute tray is a fully integrated NVLink scale-out architecture, so that each GPU in the entire rack is directly connected to every other GPU via NVLink. For compute network connectivity, each compute tray includes 2 Mezzanine Network Boards with 2 ConnectX-8 silicon chips in each, for a total of 4 ConnectX-8 Host Channel Adapters (HCA). Each tray also includes 1 M.2 NVMe device for Operating System (OS) storage and 4 E1.S NVMe storage devices typically used as a fast, local cache.

Figure 3 Sample logical design for the compute tray



To ensure balanced compute performance with the GPUs, each CPU is paired with a NVIDIA ConnectX-8 Mezzanine Board. Additionally, to support advanced storage acceleration, each tray includes a NVIDIA Bluefield-3 DPU.

Table 1 Compute Tray Components

Component	Quantity
NVIDIA Grace Processor, with a total of 72 ARM Neoverse V2 cores, connected via NVLink C2C, and 1TB aggregated LPDDR5 CPU main memory	2
NVIDIA B300 GPU, with 1,152 GB aggregated HBM3 memory	4
NVIDIA ConnectX-8 Mezzanine Boards, with 2 ConnectX-8 network adapters each	2
Dual-port QSFP112 NVIDIA BlueField-3 B3240 DPU	1

Control Plane/Management Nodes

The customer should have guidance from the OEM regarding an IaaS-managed control plane nodes. The High Availability (HA) features of the control plane nodes should be enabled to ensure reliability for the enterprise customer. In this Enterprise RA, we assume a control plane composed of 12 management nodes, providing sufficient capacity and HA for large enterprise deployments. Table 2 outlines the specifications of the x86 Management Nodes while Table 3 outlines the specifications of the ARM Management Nodes.

Table 2 Management Node Components (x86 control node)

Component	Configuration
CPUs	Intel based x86 2x Socket, 32c
System Memory	512 GB (8x DDR5 64GB)

Boot Drive	OS 2x960 GB M.2 or SATA/SAS SSD/ SW RAID1
Data Cache Drive	2x 7.68 TB NVMe SSD
Storage/in-band Network Adapter	4x ConnectX-7 200G single-port
Platform Security	TPM 2.0, SHA-256 and support for secure boot
Baseboard Management Controller (BMC)	Host BMC with discrete connection, supports Redfish 1.4 or above, Serial-over-LAN via SSH, IPv6 support, support discovery of BMC IP in-band via Redfish
Management Network	2 x 1 Gb/s In-band port
Host UEFI	UEFI 2.6 compliant BIOS implementation for HTTP(s) network boot

Table 3 Management Node Components (ARM control node)

Component	Configuration
CPU	2x NVIDIA Grace with 72 cores
System Memory	480 GB (LPDDR5X on-module)
Boot Drive	OS 2x960 GB M.2 or SATA/SAS SSD/ SW RAID1
Data Cache Drive	2x 7.68 TB NVMe SSD
Storage/in-band Network Adapter	4x ConnectX-7 200G single-port
Platform Security	TPM 2.0, SHA-256 and support for secure boot
Baseboard Management Controller (BMC)	Host BMC with discrete connection, supports Redfish 1.4 or above, Serial-over-LAN via SSH, IPv6 support, support discovery of BMC IP in-band via Redfish
Management Network	2 x 1 Gb/s In-band port
Host UEFI	UEFI 2.6 compliant BIOS implementation for HTTP(s) network boot

NVIDIA NVLink

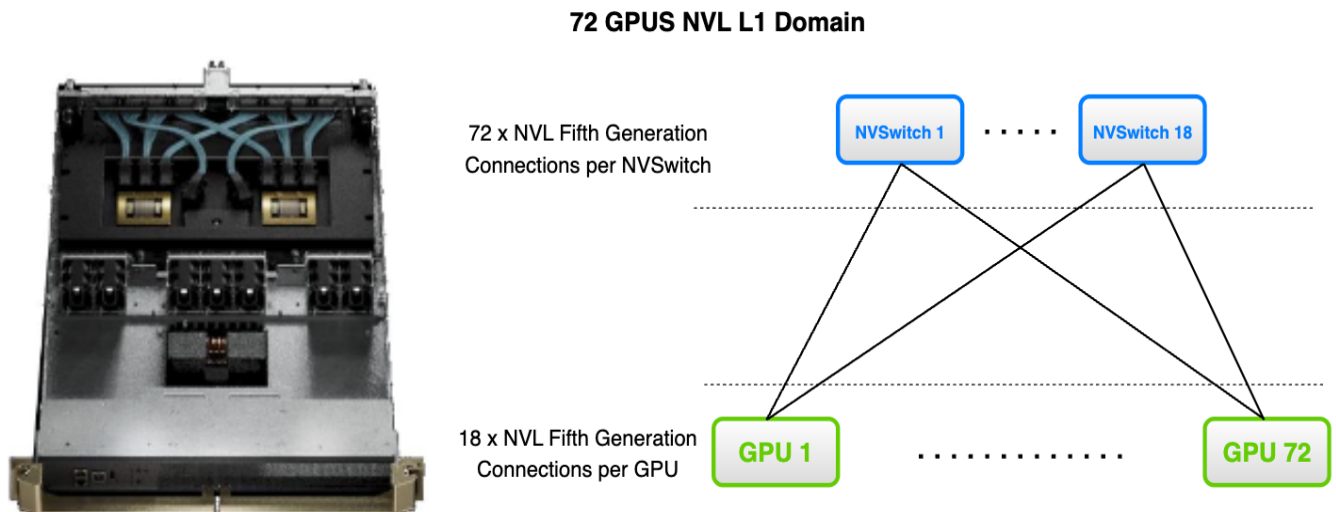
The NVIDIA Blackwell architecture introduces fifth-generation NVLink, delivering up to 1800 GB/s per GPU – doubling the bandwidth of the previous generation. A new NVSwitch chip and NVLink switch have also been introduced to enable even larger NVLink domain sizes. Together, they enable high bandwidth communication between Blackwell GPUs and unlock significant benefits for real-time, cost-effective large model inferencing.

Since the GB300 NVL72 integrates 36 NVIDIA Grace CPUs with 72 NVIDIA Blackwell Ultra GPUs in a rack-scale design, the fifth-generation NVLink enables all 72 GPUs to function as a single multi-GPU unit of compute enabling 30x faster real-time trillion-parameter inference compared to prior generations.

NVIDIA NVLink Switch Tray

Each GB300 NVL72 rack includes 9 NVLink Fifth-Generation switch trays, with 2 NVSwitch ASICs per tray. Each GPU has 18 NVLink Fifth-Generation links, one per in-rack NVSwitch via the copper backplane. This way a single rack may form a fully connected L1 Domain with 72 GPUs. The NVLink Switch is a managed switch running the NVOS networking operating system.

Figure 4 NVLink Switch



NVIDIA GB300 NVL72 Networking

The NVIDIA GB300 NVL72 platform networking configuration enables tremendous AI performance and scale. It leverages the NVIDIA expertise in AI cloud data centers and optimizes network traffic flow:

- **East/West East/West(Compute Network) traffic:** This refers to traffic between the compute nodes/trays within the GB300 NVL72 cluster, typically for multi-node AI training, HPC collective operations, and other workloads.
- **North/South (Customer and Storage Network) traffic:** This involves traffic between the GB300 NVL72 and any external resources including cloud management and orchestration systems, remote data storage nodes, and other parts of the data center or the Internet.

The GB300 NVL72 comes pre-integrated with a comprehensive networking stack design to optimize East/West and North/South traffic. Each compute tray is equipped with ConnectX-8 network interface cards (NICs) delivering 800 Gb/s and BlueField-3 data processing units, delivering up to 400 Gb/s. This configuration ensures high-bandwidth, low-latency communication across the 18 compute trays and during data ingress and egress from the system. This architecture simplifies deployment and ensures consistent performance across the system.

Compute (Node East/West) Ethernet Networking

Each tray within the GB300 NVL72 rack has four dual-port ConnectX-8 SuperNICs for GPU to GPU (East/West) traffic. Each ConnectX-8 SuperNIC offers up to 800 Gb/s, low-latency network connectivity between GPUs in the AI cluster. ConnectX-8 SuperNIC offers up to 800 Gb/s, low-latency network connectivity between GPUs in the AI cluster, featuring RDMA and RoCE acceleration, with NVIDIA® GPUDirect® and GPUDirect Storage technologies. For data centers that deploy Ethernet, ConnectX-8 offers a range of advancements including RoCE optimizations and multi-tenancy, which are key to in AI cloud environments. Additionally, ConnectX-8 offers advanced hardware offloads for overlay networks such as VXLAN and NVGRE, which help reduce CPU load and improve network efficiency.

Beyond raw speed, ConnectX-8 enhances network performance with features like advanced congestion control, telemetry-based routing, and quality of service (QoS) capabilities to meet diverse AI application and training requirements. Security is built in with hardware acceleration for cryptographic protocols like IPsec and MACsec, taking these tasks over from the CPU to maintain performance.

Multi-node deployments with an NVIDIA B300 platform should adhere to the following total compute network bandwidth per GPU recommendations:

The ConnectX-8 SuperNIC is integrated onto the baseboard, maintaining a **1:1 GPU-to-NIC ratio** to ensure optimal bandwidth and direct connectivity for each GPU.

Total Minimum Compute Network Bandwidth

- 400 GB/s (8x 400 Gb/s Ethernet NICs)

Total Recommended Compute Network Bandwidth

- 800 GB/s (16x 400 Gb/s Ethernet NICs using breakout)

Converged (Node North/South) Ethernet Networking

This section describes the BlueField-3 role for the North/South (North/South) Ethernet network in the GB300 NVL72. Each tray within the GB300 NVL72 rack has one dual-port NVIDIA BlueField-3 B3240 DPUs for North/South traffic. The NVIDIA BlueField-3 B3240 is a PCIe based data processing unit (DPU) that can handle an aggregate bandwidth of approximately 480 Gb/s. The BlueField-3 DPU is optimized for the North/South network

such as storage, complementing the ConnectX-8 SuperNIC which is optimized for the East/West, AI compute fabric.

The BlueField-3 B3240 DPU offers several essential capabilities and benefits within the NVIDIA GB300 NVL72 platform:

- **Out of Band Provisioning:** BlueField-3 serves as an optimized compute platform for the data center control-plane, enabling automated provisioning and elasticity. This empowers GB300 NVL72 to scale resources dynamically based on fluctuating demand, ensuring efficient allocation of computing resources for transient AI workloads.
- **Storage Acceleration:** BlueField-3 provides advanced storage acceleration features that optimize data storage access. Its innovative BlueField SNAP technology enables remote storage devices to function as local, improving AI performance and streamlining cloud operations.
- **Secure Infrastructure:** BlueField-3 operates in a highly secure zero-trust mode and functions independently from the host, significantly enhancing the security of the GB300 NVL72 platform. In addition, BlueField-3 enables a wide range of accelerated security services, including next-generation firewall and micro-segmentation, bolstering the overall security posture of the infrastructure.

BlueField-3 B3240 DPU's integration in the NVIDIA GB300 NVL72 platform delivers crucial capabilities that enhance resource management, performance, security, and scalability, making the NVIDIA GB300 NVL72 platform an ideal choice for AI workloads in datacenter environments.

The BlueField-3 DPU supports three modes of operation, with the Embedded Function (ECPF) or DPU mode used in this Enterprise RA. In this mode, the NIC resources and functionality are owned and controlled by the embedded Arm subsystem. All network communication to the host flows through a virtual switch control plane hosted on the DPU Arm cores, and only then proceeds to the host.

To securely control and manage the NVIDIA GB300 NVL72 platform through the BlueField DPU, NVIDIA has integrated an onboard Baseboard Management Controller (BMC) into its BlueField products. The onboard BlueField BMC enables the provisioning and management of the BlueField and NVIDIA GB300 NVL72 using standard tools, including Redfish APIs. It includes an external root-of-trust to ensure that the BMC firmware is secured. The main interface of the BlueField BMC is a 1Gb/s out-of-band management port that connects to the data center and management network.

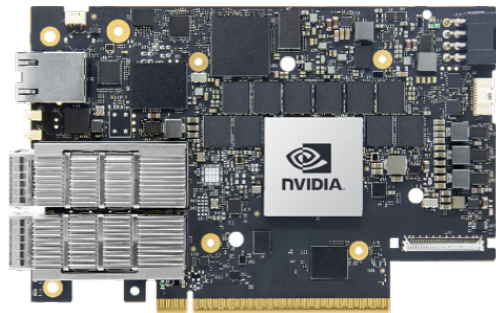
Networking Hardware

This section describes the networking hardware used to connect components of the Enterprise RA

NVIDIA BlueField-3 B3240 DPU

The [NVIDIA BlueField-3 B3240 DPU](#) with two 400 Gb/s ports (Figure 5) is used in the in-band management and storage fabric. Even though the B3240 DPU supports 400Gb/s per port, the card only supports an aggregate bandwidth across both its ports of approximately 480Gb/s. The Enterprise RA recommends the use of dual ports for the best performance and flexibility. Each compute tray has one B3240 DPUs.

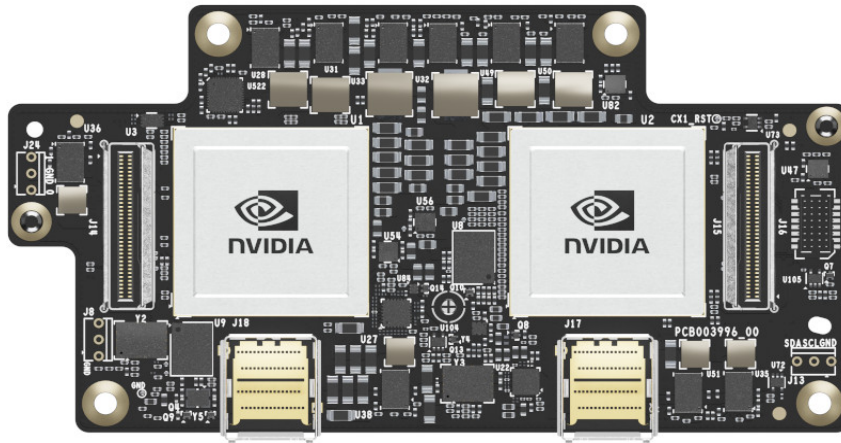
Figure 5 NVIDIA Bluefield-3 B3240 DPU



NVIDIA Mezzanine Network Board with ConnectX-8

Each GB300 compute tray has a Mezzanine Network Board (**Figure 6**) that is used for Compute (East/West) network. The tray has two such boards, with two ConnectX-8 devices on each board.

Figure 6 NVIDIA Mezzanine Network Board with two ConnectX-8 ASICs



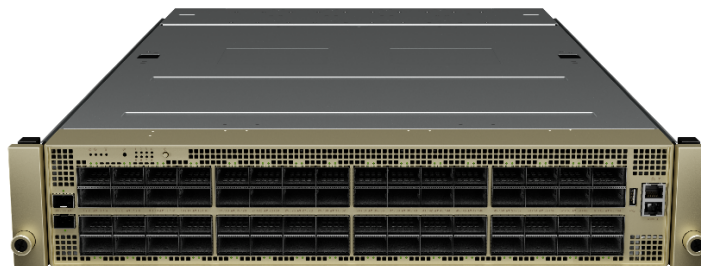
NVIDIA Ethernet Interconnects

The Enterprise RA configurations must be installed with NVIDIA Ethernet interconnects. NVIDIA Ethernet interconnects make extensive use of pluggable optical transceivers and detachable optical fibers for easier installation, inspection, and debugging.

NVIDIA SN5610 Switch

The [NVIDIA SN5610 switch](#) (Figure 7) both offer 64 total ports of 800 Gbps to provide connectivity for Compute (East/West), Customer and Storage (North/South), in-band management, and storage in the Enterprise RA.

Figure 7. NVIDIA SN5610 Switch



The NVIDIA SN5610 adds two SFP28 ports and makes switch testing easier since the ports are in pairs.

NVIDIA SN2201 Switch

The [NVIDIA SN2201 switch](#) (Figure 8) offers 48 ports to provide connectivity for out-of-band (OOB) management. OOB management provides consolidated management connectivity for all components in the Enterprise RA.

Figure 8. NVIDIA SN2201 Ethernet switch



Networking Physical Topologies

The Enterprise RA configurations use three physical network fabrics:

- GPU Compute (Node East/West) Network
- CPU Converged (Node North/South) Network
- Out-of-Band Management Network

Each network is discussed in this section.

GPU Compute (Node East/West) Network

The compute fabric (East-West) is built using switches with NVIDIA Spectrum technology in a full non-blocking fat tree topology to provide the highest level of performance for applications running over the GB300 NVL72 cluster. The compute fabric is an RDMA-based fabric. It is designed to provide the shortest hops through the network for the application, in a leaf-spine manner, where the GPUs are connected using a rail-optimized network topology through their respective ConnectX-8 NICs. This design allows the most efficient communication for multi-GPU applications within and across the trays.

The leaf and spine architecture allows for a scalable and reliable network that can fit varied sizes of clusters using the same architecture.

The compute fabric is designed to facilitate redundancy, maximize bandwidth and minimize network latency required to connect GPUs within a server and within a rail. Models beyond 1920B parameters (assuming FP4 quantization) would require model parallelism and more than a single tray of the GB300 NVL72 rack, which are connected through NVLink and NVSwitch technology.

Multi-Plane Topology Approach

While each GPU is associated with a dedicated ConnectX-8 SuperNIC, the coupling of a single link from a NIC to a single point of failure (SPOF) in the fabric is a concern. Customers are opting for a load balanced multi-link approach which can satisfy both the full capacity of the GPU bandwidth as well as avoiding a SPOF. Each compute networking plane forms a separate fabric, where the resiliency and the load balancing between the two planes is handled by the NCCL on the host. Any failure to provide GPU to GPU connectivity via one plane will seclude traffic to the alternative plane, while if both are active, the traffic will be balanced between planes in a manner that utilizes the bandwidth of both links. Load balancing via NCCL is indicated as more efficient than other L2/L3 bonds like LAG that only have “local” awareness for traffic balancing and static hash-based distribution which is suboptimal for AI workloads due to low entropy.

A second attribute for planes separation is the expansion of the fan-out of the NIC and switches. A higher radix of interfaces enables building a larger network with fewer network tiers in a Clos topology.

Dual Plane Topology

In this Enterprise RA we utilize a Dual Plane topology. A Dual Plane Topology involves building two identical planes that connect to each GPU interface. With each GPU generating 800 Gb/s bandwidth through the ConnectX-8 SuperNICs, dual plane topology involves breaking the interface to 2x400 Gb/s interfaces. Every such interface is then connected to a different leaf switch, and every such leaf switch is part of an independent fabric that scales to 1024 interfaces of 400 Gb/s as part of this reference architecture. The GPU is not aware of the two independent fabrics, each carrying 50% of the generated traffic, and is only aware that 100% of traffic is carried by the ConnectX-8 SuperNIC. Tracking of each plane, load balancing, and failure handling is handled by the ConnectX-8 SuperNIC on the hardware level. A failing or degraded plane will carry an impact linearly associated with the dropped bandwidth.

CPU Converged (Node North/South) Network

A converged network for both storage and in-band management is used in the Enterprise RA. This provides Enterprises with flexible storage allocation and easy network management. The converged network has the following attributes:

- > Provide high bandwidth to shared storage and connects the customer through the converged network.
- > It is independent of the compute fabric to maximize both storage and application performance.
- > Each compute tray connects to two separate switches using dual 400 Gb/s ports, while each management connects to the same switches with four 200 Gb/s ports. This design provides redundancy and enables per-node storage bandwidth of up to 40 GB/s.
- > The fabric is built on Ethernet technology with RDMA over Converged Ethernet (RoCE) support and utilizes NVIDIA BlueField-3 B3240 DPUs in each compute tray to deliver existing and emerging cloud and storage services.
- > It is flexible and can scale to meet specific capacity and bandwidth requirements.
- > Hybrid storage fabric design with support for tenant isolation provides access to both shared and dedicated storage per tenant.
- > Used for node provisioning, data movement, Internet access, and other services that must be accessible by the users.

Out-of-Band Management (Node) Network

The OOB management network connects all the base management controller (BMC) ports, as well as other devices that should be physically isolated from system users to allow the infrastructure management. This includes the 1 Gb/s switch management ports and the BlueField-3 DPU management ports.

Network Logical Architecture

This Enterprise RA uses a spine-leaf network providing physical fabrics for the use-cases and requirements shown in Table 4:

Table 4 Fabric Roles and Usage Recommendations

Fabric Roles	RDMA/RoCE Support	AI Training, Fine tuning, ML or HPC	Pure Inference	Storage connectivity
GPU Compute (East/West)	Recommended	Recommended	Optional	Not Recommended
CPU Converged (North/South)	Optional	Recommended	Recommended	Recommended
Storage	Optional	Recommended	Recommended	Recommended
Customer	Not applicable	Recommended	Recommended	Optional
Support Servers	Optional	Recommended	Recommended	Optional
OOB Management	Not applicable	Recommended	Recommended	Not Applicable

Deeper explanations of each of the fabric roles are below:

- GPU Compute (East/West) Network

The GPU Compute (East/West) network is an RDMA based fabric in a leaf-spine architecture where the GPUs are connected using a rail-optimized network topology through their respective SuperNICs. This design allows the most efficient communication for multi-GPU applications within and across the compute nodes. In the architectures below for smaller design points the Compute Fabric has been merged with the Converged Fabric. This is to lower cost and to simplify the network architecture.

For larger network design points due to the high number of endpoints required the Compute Fabric utilizes a super-spine, spine and leaf networking architecture. This is principally to support the required design points but also to allow extensibility in the architecture.

- CPU Converged (North/South) Network

The CPU Converged (North/South) network connects the nodes using two 400 Gb/s ports to two separate switches to provide redundancy and high storage throughput. This network is used for node communications with compute, storage, in-band management, and end-user connections.

- Storage connectivity

This network specifically provides converged connectivity for storage infrastructure. Principally storage is attached on this network and then can be provided to the trays/nodes via the CPU Compute network as required.

- Customer Network Connectivity

Typically, this is upstream connectivity to connect the cluster into the rest of an Enterprise customers networking infrastructure. This is provisioned to provide ample bandwidth for typical user communications.

- Support Server Networking

This is dedicated networking for the respective support servers. These servers generally provide management, provisioning, monitoring, and control services to the rest of the cluster. High performance networking is used here due to requirements such as cluster deployment and imaging.

- Out-of-band Management Networking for the infrastructure

All infrastructure requires management. This network provides bulk management 1Gb RJ45 connectivity for all the nodes. This network uses low-cost bulk management switches. These switches have upstream connections to the Core networking infrastructure to allow flexibility and wider consolidation of services such as management and monitoring.



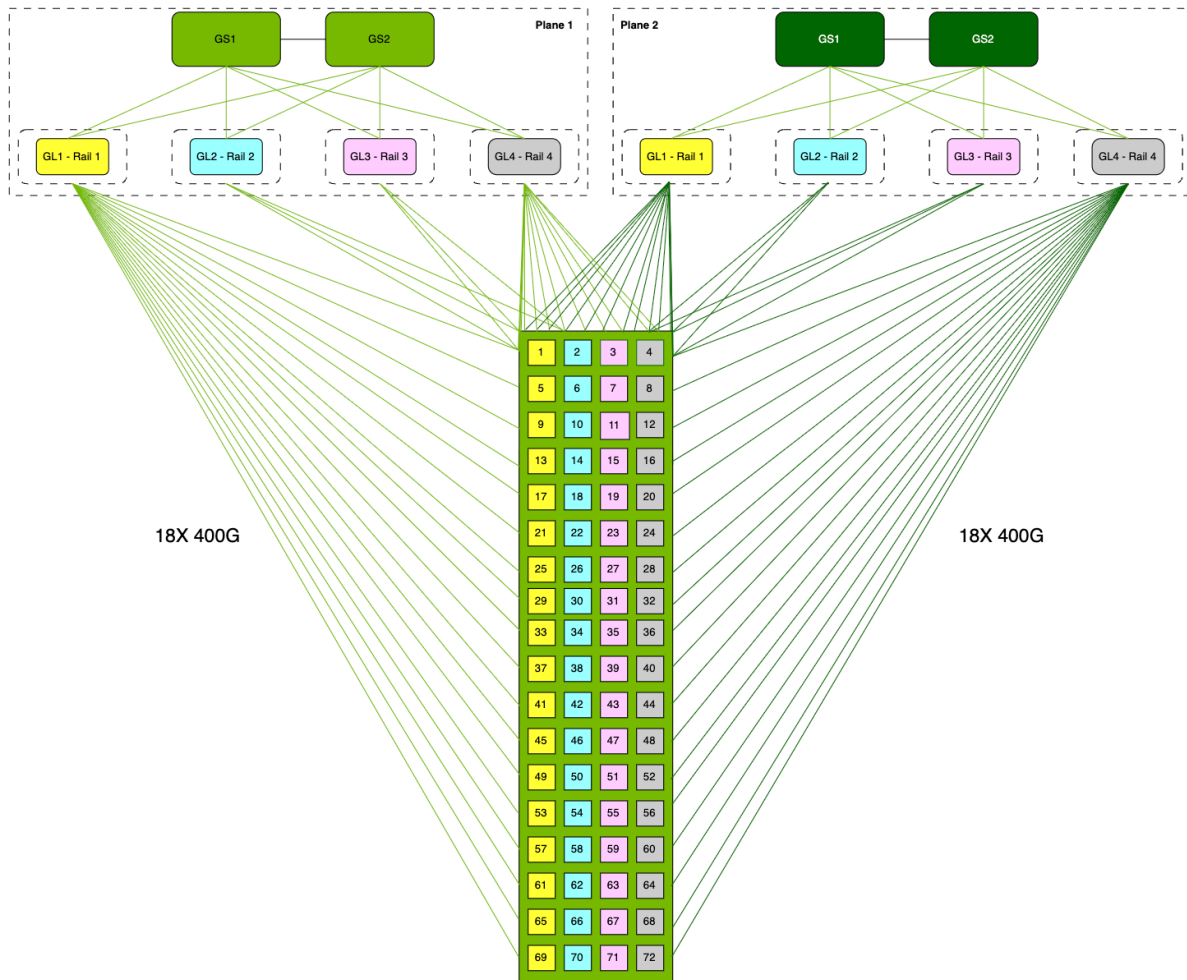
Note: VLAN isolation would be used to provide logical separation of the networks over the single physical fabric.

The exception to the above architecture is for the smaller cluster layouts where a collapsed Spine-Leaf design has been used.

Enterprise RA Scalable Unit (SU)

The Enterprise RA is built on scalable units (SU) based on 1 rack of GB300 NVL72 which contains 18 compute nodes (trays). Each SU is a discrete entity of computation that is tied to the port availability size of the network devices. SUs can be replicated to adjust the scale of the deployment with more ease.

Figure 10 Example diagram of 18-Tray SU connected to 2 planes (4 rails each) of the compute fabric



The single rack scalable unit provides the following connectivity building blocks:
Within the rack: As shown in

- Figure 10 all 72 GPUs are interconnected in a single NVLink domain, allowing them to function as a single multi-GPU unit of compute with a bandwidth of 900GB/s (1800 GB/s bi-directional)
- For the Compute (East/West) fabric: 18 trays, each with 4 x single-port NVIDIA ConnectX-8 NICs and a total aggregate bandwidth of 3200 Gb/s
- For the Converged (North/South) fabric: 18 trays, each with 1x B3240 DPU providing 2x 400Gb/s connections and a total aggregate bandwidth of 800 Gb/s
- For the Out-of-band Management fabric, 18 trays, each with 3x 1Gb/s connections providing 54 x 1Gb/s for management

Spine-Leaf Networking

The network fabrics are built using switches with NVIDIA Spectrum-X Ethernet technology in a full nonblocking fat tree topology to provide the highest level of performance for the application running over the Enterprise RA configuration. The networks are RDMA compliant based fabrics in a leaf-spine architecture where the GPUs are connected using a rail-optimized network topology through their respective SuperNICs. This design allows the most efficient communication for multi-GPU applications within and across the nodes.

The leaf and spine architecture allows for a scalable and reliable network that can fit varied sizes of clusters using the same architecture. The compute network is designed to maximize bandwidth and minimize network latency required to connect GPUs within a server and within a rail.

In addition to the supported compute fabrics the converged spine-leaf network also supports the following attributes:

- Provides high bandwidth to high performance storage and connects to the data center network.
- Each compute and management node are connected with two 400 Gb/s ports to two separate switches to provide redundancy and high storage throughput.

Out-of-Band (OOB) Management Network

The OOB management network connects the following components:

- Base management controller (BMC) ports of the server nodes
- BMC ports of the Bluefield-3 DPU
- OOB management ports of switches

This OOB network can also connect other devices that should have management connectivity physically isolated for security purposes. The SN2201 is used to connect to the BMC/OOB 1 Gb/s ports of these components. Scale out this network using a 25 Gb/s or 100 Gb/s spine layer to connect the SN2201 uplink switches.

Optimized Use Case

This architecture is optimized for both inference and training use cases. Deployment of this single architecture will enable both types of workloads.

Architecture Details

The components of the Enterprise GB300 NVL72 Enterprise RA using Spectrum-X 800 Gbps for the compute fabric are described in Table 5.

Table 5 GB300 NVL72 Server Ethernet Components with Spectrum-X Fabric

Component	Technology
Compute node servers (18)	Each GB300 NVL Compute Tray is configured with • One NVIDIA BlueField-3 B3240 dual port 400 Gb/s DPU. • Four NVIDIA ConnectX-8 SuperNICs dual-port 800 Gb/s. The adapters operate at 2x400 Gb/s per port
Compute (East/West) Spine-Leaf Fabric	NVIDIA SN5600 128-port 400 Gb/s switches
Converged (North/South) Spine-Leaf Fabric	NVIDIA SN5600 128-port 400 Gb/s switches
OOB management fabric	NVIDIA SN2201 48-port 1 Gb plus 4-port 100 Gb/s

Compute (Node East/West) Fabric Table

Table 6 below shows the number of cables and switches required for the **dual-plane** Compute (Node East/West) Fabric for different SU sizes.

Table 6 Compute Node (Node East/West) and Switch Component Count

Compute Counts			Switch Counts			Transceiver Counts			Cable Counts	
Nodes	GPUs	SUs	Leaf	Spine	Upinks per leaf to	Node to Leaf		Switch-to -switch	Compute- to -Leaf	Switch-to -switch
						Compute	Switch			
36	144	2	8	4	18	144	144	288	288	288
72	288	4	16	8	9	288	288	576	576	576
144	576	8	32	12	6	576	576	1152	1152	1152

Converged (Node North/South) Fabric Table

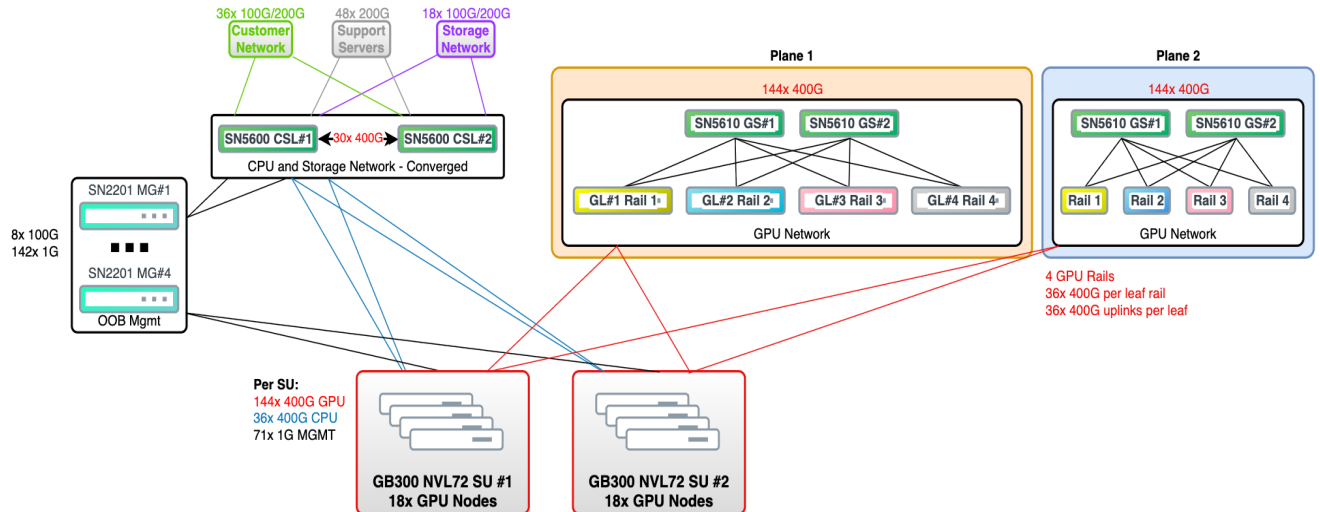
Table 7 below shows the number of cables and switches required for the Converged (Node North/South) Fabric for different SU sizes.

Table 7 Converged (Node North/South) and Switch Component Count

Compute Counts			Switch Counts		Converged Network Allocated Ports							Transceiver Counts			Cable Counts		
Nodes	GPUs	SUs	Leaf	Spine	(Node N-S)	Storage	Mgmt Uplinks	Customer	Support	(both ends)	Endpoint to Leaf			Switch-to -switch	Node-to- Leaf	Other-to- Leaf	Switch-to -switch
											Node	Other	Switch				
36	144	2	2	N/A	36	8	4	12	12	30	72	102	58	40	72	40	32
72	288	4	4	2	72	12	4	20	12	256	144	156	100	274	144	52	260
144	576	8	7	4	144	20	8	36	12	448	288	264	184	484	288	78	456

2 Racks, 36 Trays with 144 Blackwell Ultra GPUs

Figure 11 Two Rack GB300 NVL72 with SpectrumX Enterprise RA (144 GPUs)



Architecture Overview

- Two GB300 NVL72 scalable units (SUs) composed of 36 server nodes total (18 nodes per SU)
- 144 NVIDIA Blackwell GPUs connected through the Spectrum-X compute fabric
- Each rack requires 2x SN5600 switches for CPU and storage connectivity and up to 12x SN5600 switches for the dual-plane GPU network
- Infrastructure supports scaling from one to two SUs; leaf switches are intentionally underpopulated to align with GB300 NVL72 software design criteria

Network Design

- Cost-optimized collapsed spine-leaf architecture for all required fabrics
- Spectrum fabric implemented as a rail-optimized, non-blocking 64-port switch design
- Port breakout used to consolidate connections without reducing resiliency
- Each leaf switch is capable of serving three racks but is configured for two to maintain balance and performance

Connectivity (Under Optimal Conditions)

- 144x 400G uplinks per SU for GPU Compute traffic
- 36x 400G uplinks per SU for CPU fabric
- Up to 12 support server nodes, quad-connected at 200G

- 36x 100G/200G connections to the customer network (minimum 25Gb bandwidth per GPU)
- 18x 100G/200G storage connections (minimum 12.5Gb bandwidth per GPU)

Compute Fabric Design

- Dual-plane Spectrum fabric ensures high availability, resiliency, and balanced load distribution
- Each plane uses 4 leaf switches, each assigned to the same GPU position across all trays (for example, GPU 1 [Yellow], GPU 2 [Blue], GPU 3 [Pink], GPU 4 [Grey])
- Plane 2 mirrors Plane 1 to provide redundancy and throughput stability
- ConnectX-8 800Gb ports are split into 2x 400Gb links; each GPU has two 400Gb paths (one to each plane)
- Dual-ported optics are recommended to simplify breakout at the ConnectX-8 OSFP port

Management

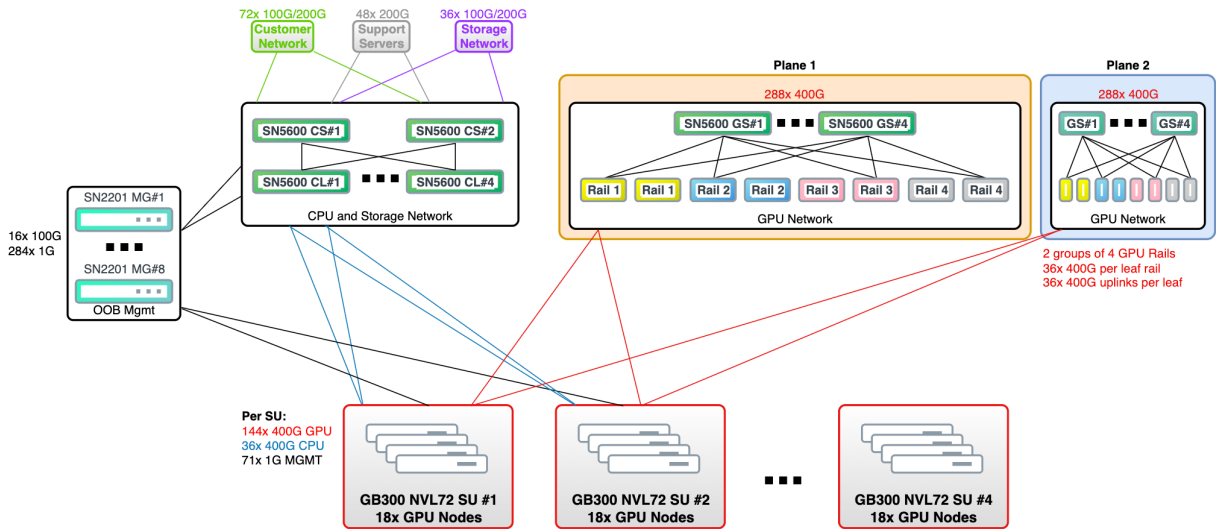
- 100Gb/s management fabric with high availability using two 100Gb/s connections per switch to the core network
- All nodes connected to 1Gb management switches, which are uplinked into the core management network
- VLAN isolation is used to segregate the various north-south and east-west fabrics within the collapsed physical infrastructure

Additional Considerations

- Rack layouts are assumed to provide power redundancy within each cabinet; if not available, an alternative rack layout should be considered
- Under-populated port strategy and breakout usage preserve headroom for future scalability while maintaining operational resilience

4 Racks, 72 Trays with 288 Blackwell Ultra GPUs

Figure 12 Four Rack GB300 NVL72 with Spectrum-X Enterprise RA (288 GPUs)



Architecture Overview

- Four GB300 NVL72 scalable units (SUs) with 72 trays/nodes and 288 Blackwell Ultra GPUs interconnected via the Spectrum-X compute fabric
- Networking sized to scale between three and four SUs while keeping a consistent per-SU node count of 18 trays
- Design references switch, transceiver, and cable requirements as summarized in the associated bill-of-materials tables (Table 6 and Table 7)

Network Design

- Cost-efficient collapsed spine-leaf architecture for all fabrics except the GPU Compute (E-W) network
- GPU Compute (E-W) fabric implemented as a separate, isolated leaf-spine network to deliver high-bandwidth, low-latency node-to-node communication
- Port breakout used wherever possible to consolidate ports without compromising resiliency
- Leaf switches can support up to three racks but are intentionally underpopulated to serve only two racks to satisfy GB300 NVL72 software requirements

Connectivity (Under Optimal Conditions)

- 144x 400G uplinks per SU for GPU Compute traffic
- 36x 400G uplinks per SU for CPU traffic
- Up to 12 support server nodes, all quad-connected at 200Gb
- 72x 100G/200G connections toward the customer network (minimum 25Gb bandwidth per GPU)
- 36x 100G/200G storage connections (minimum 12.5Gb bandwidth per GPU)

Compute Fabric Design

- Spectrum compute fabric for the Node E-W network is rail-optimized and non-blocking using 64-port switches in a full spine-leaf topology
- Dual-plane Spectrum fabric provides high availability, resiliency, and load-balancing for GPU traffic
- Each plane has 4 leaf switches; each leaf is wired to the same GPU position across all trays (for example, GPU 1 [Yellow], GPU 2 [Blue], GPU 3 [Pink], GPU 4 [Grey])
- Plane 2 mirrors Plane 1, and each ConnectX-8 800Gb port is broken out into 2x 400Gb links so each GPU has a dedicated 400Gb connection to each plane
- Dual-ported optics at the ConnectX-8 OSFP port are recommended to simplify breakout, as detailed in NVIDIA's optical module documentation

Management

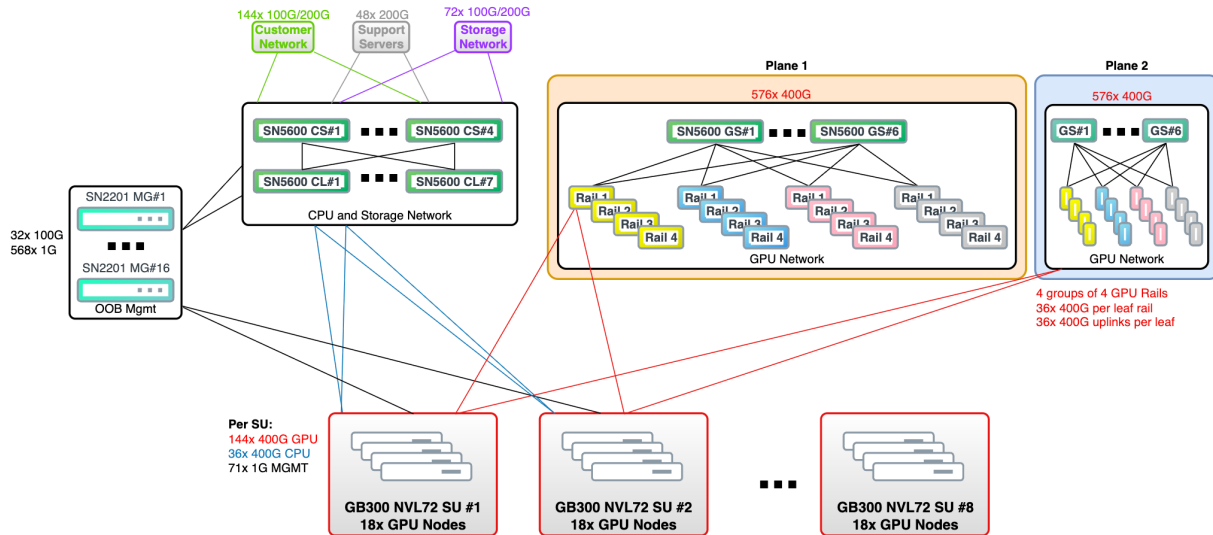
- 100Gb/s management fabric with high availability using two 100Gb/s connections per management switch to the core network
- All nodes connect to 1Gb management switches, which are then uplinked into the core network
- Per-SU management uses two SN2201 switches; in a four-rack design this yields eight SN2201 switches, each uplinked to the core via 2x 100G links

Additional Considerations

- VLAN isolation is applied across the converged Node North-South network fabrics that share the underlying physical infrastructure
- Rack layout is assumed to provide power supply redundancy within each rack; if this cannot be met, an alternative rack layout should be considered

8 Racks, 144 Trays with 576 Blackwell Ultra GPUs

Figure 13 Eight Rack GB300 NVL72 with Spectrum-X Enterprise RA (576 GPUs)



Architecture Overview

- Eight GB300 NVL72 scalable units (SUs) with 144 trays/nodes and 576 Blackwell GPUs interconnected via Spectrum-X compute fabric.
- Networking sized to scale between seven and eight SUs while keeping 18 trays per SU.
- Switch, transceiver, and cable requirements are defined in the associated bill-of-materials tables (Table 6 and Table 7).

Network Design

- Cost-efficient spine-leaf architecture for all fabrics except the GPU Compute (E-W) fabric.
- GPU Compute (E-W) network implemented as a separate, isolated leaf-spine fabric to deliver high-bandwidth, low-latency node-to-node communication.
- For very large deployments (around 1024 nodes and above), a super-spine layer is introduced to maintain non-blocking point-to-point connectivity.
- Port breakout is used wherever possible to consolidate ports while preserving resiliency.
- Each leaf switch can support up to three racks but is intentionally underpopulated to serve only two racks to meet GB300 NVL72 software requirements.

Connectivity (Under Optimal Conditions)

- 144x 400G uplinks per SU for GPU Compute traffic.

- 36x 400G uplinks per SU for CPU traffic.
- Up to 12 support server nodes, all quad-connected at 200Gb.
- 144x 100G/200G connections toward the customer network (minimum 25Gb bandwidth per GPU).
- 72x 100G/200G storage connections (minimum 12.5Gb bandwidth per GPU).

Compute Fabric Design

- Spectrum fabric for the Node E-W network is rail-optimized and non-blocking using 64-port switches in a full super-spine–spine-leaf topology.
- Dual-plane Spectrum fabric provides high availability, resiliency, and load-balancing for GPU traffic.
- Each plane uses 4 leaf switches; each leaf is wired to the same GPU position across all trays (for example, GPU 1 [Yellow], GPU 2 [Blue], GPU 3 [Pink], GPU 4 [Grey]).
- Plane 2 mirrors Plane 1, and each ConnectX-8 800Gb port is broken out into 2x 400Gb connections, so each GPU has a dedicated 400Gb link to each plane.
- NVIDIA recommends using a dual-ported optic at the ConnectX-8 OSFP port to simplify breakout, as described in the referenced optic documentation.

Management

- 100Gb/s management fabric with high availability using two 100Gb/s connections per management switch to the core network.
- All nodes are connected to 1Gb management switches, which are then uplinked into the core network.
- Per-SU management uses two SN2201 switches; in an eight-rack design this yields 16 SN2201 switches, each uplinked to the core via 2x 100G links.

Additional Considerations

- VLAN isolation is used across the converged North-South network fabrics that share the underlying physical infrastructure.
- Rack layout is assumed to provide power supply redundancy within each rack; if this is not available, an alternative rack layout should be considered.

Software

Infrastructure software is a critical component of the NVIDIA Enterprise Reference Architecture, providing the foundation for deploying, managing, and orchestrating AI workloads at scale. The included software (Table 8) is optimized for AI from top to bottom, with every part of the stack designed to maximize the performance and value of Enterprise RA.

Table 8 Enterprise RA Software Components

Component	Description
NVIDIA AI Enterprise	A suite of AI and data analytics software optimized for the development and deployment of AI. Provides best-in-class development tools, production-ready containers, and frameworks for rapid deployment of AI workloads
NVIDIA Mission Control	NVIDIA Mission Control is a software solution that automates and manages every aspect of AI data center operations, from infrastructure provisioning to developer workloads. It streamlines workload deployment, orchestrates jobs, and monitors resources.
NVIDIA Dynamo	NVIDIA Dynamo is an open-source inference software for efficiently orchestrating and coordinating AI inference requests across large fleets of GPUs, scaling reasoning models in AI Factories at the lowest cost and highest efficiency.
NVIDIA Run:ai	An orchestration platform that accelerates AI operations with dynamic orchestration across the AI lifecycle.
NVIDIA NGC	A collection of GPU-optimized containers for AI and HPC.
NVIDIA NetQ™	A highly scalable, modern network operations toolset that provides visibility, troubleshooting, and validation of network fabrics in real time



Note: Management software for storage is provided by storage partners and is not part of the Enterprise RA software stack.

Services

A successful deployment requires more than advanced infrastructure — it also requires comprehensive services that accelerate deployment, ensure reliability, and achieve optimal performance.

NVIDIA works closely with its global partners to extend service reach, scale delivery, and provide deep industry expertise across all stages of implementation and operation.

- **Roadmap services** – Conduct current infrastructure assessments, model future capacity scenarios and develop phased plans to guide design, build, migration and operations of data centers over time
- **First project implementation services** – Execute the initial deployment (from installation, configuration and commissioning to operational hand-over) of a first data center build or module
- **Managed services** – Operate the ongoing monitoring, maintenance, optimization of the data center including 24/7 support, contract management and vendor coordination.
- **Support Services** – Minimize downtime by providing expert guidance and access to NVIDIA specialists across compute, software and networking, ensuring optimal performance and availability.
- **Education Services** – Deliver online and hands-on training to customers and partners, providing in-depth knowledge that enables customers to maximize the value of their AI investments and partners to build deep AI expertise.

For additional information about NVIDIA Enterprise Services, visit <https://www.nvidia.com/en-us/support/enterprise/services/>.

Summary

NVIDIA Enterprise RAs are the leading data center scale architecture designed to meet the demanding and growing needs of AI factories. This Enterprise RA represents the architecture used by NVIDIA for internal AI model training and HPC research and development. The 2-4-5-800 networking pattern outlined in this document is designed for the NVIDIA GB300 NVL72 rack solution. It is interconnected using Spectrum-X for compute and converged connectivity.

This Enterprise RA represents a complete system of hardware and software necessary to implement data centers for high-performance AI functionality. The combination of all these elements keeps systems running reliably, with maximum performance, and enables users to push the bounds of state-of-the-art. The Enterprise RA includes guidance for both dual plane and single plane network topologies to suit a variety of deployment scenarios.

Appendix A – Terminology

Table 9 Terminology

Term	Definition
Node	The distinct hardware tray in the GB300 NVL72 rack that supports a bare-metal OS instance
Rack	A physical arrangement of equipment, such as servers or switches in a vertical closet
Open Compute Project (OCP) Rack	Rack based on the Open Rack V3 standard from the Open Compute Project, with the vertical space divided up into 48mm units (OU) and with power distributed using power shelves, a DC bus bar and, AC power cords
Electronic Industries Alliance (EIA) Rack	Conventional “Enterprise” Rack, originally based on the EIA-310-D standard, with the vertical space divided up in 44.45mm rack units (RU) and with power distributed using rack power distribution units (rPDUs) and AC power
Tray	One of the systems inserted into the GB300 NVL72 solution, generally either a “Node” or a “NVSwitch” (which may include 1-2 NVSwitch ASICs) installed directly into a Rack
Hot Air Containment (HAC)	Refers to a set of racks that exhaust into the same hot-air containment area, generally from two opposing rows
NVLink (?) Switch ASIC	The silicon building block with a specific logical port count that drives the network topology
Global Fabric Manager (GFM)	The external services that manage the fabric of a NVLink Domain
NVLink Domain	The full set of nodes connected with a single multi-node NVLink fabric
NVLink Block	A group of nodes within a NVLink Domain that are allocated to the same job. A single job may have multiple blocks assigned across multiple domains
NVLink Partition	A construct of group nodes within an NVLink Domain that are isolated to only access each other’s GPU memory. A partition could be configured as an entire domain or there could be multiple partitions in the same domain. Using partitions should protect interruptions to work running on one partition from the other partitions. Partitions are setup via API calls to the GFM.

Appendix B – Node Configurations

This appendix provides information on the compute trays and networking configurations.

Compute Trays

Table 10 shows the GB300 compute tray requirements

Table 10 GB300 compute node/tray requirements

Component	Configuration
CPUs	2x NVIDIA Grace (C2), with total of 72 ARM Neoverse V2 cores connected with NVLink C2C
System Memory	1 TB aggregated CPU LPDDR5 Main Memory
Boot Drive	1 x 2 TB Gen4 M.2 for OS SSD
Data Cache Drive	8 x 4 TB E1.S for Data Cache SSD
NVIDIA GPUs	4x NVIDIA Blackwell Ultra GPU, with total of 720 GB of aggregated HBM3 memory
Compute Network Adapter	2x NVIDIA Mezzanine Boards with 2x ConnectX-8 800 GB/s NICs in each one
Storage/in-band Network Adapter	1x Dual-port QSFP112 NVIDIA BlueField-3 DPU operating at 400 GB/s
Platform Security	TPM 2.0, SHA-256 and support for secure boot
Baseboard Management Controller (BMC)	Host BMC with discrete connection, supports Redfish 1.4 or above, Serial-over-LAN via SSH, IPv6 support, support discovery of BMC IP in-band via Redfish
Host UEFI	UEFI 2.6 compliant BIOS implementation for HTTP(s) network boot

Switches and Cables

Table 11 and Table 12 provides an overview of the switches and cables used in the cluster. The network adapters of the GB300 NVL72, the storage servers, and the control plane servers are included for better understanding of the connectivity requirements.

Table 11 Switching with required inter-switch links, transceivers, and cables

Component	Scalable Units (SUs)							
	1	2	3	4	5	6	7	8
GB300 NVL72 Racks	1	2	3	4	5	6	7	8
GB300 NVL72 Compute Trays	18	36	54	72	90	108	126	144
Grace Processors (two per tray)	36	72	108	144	180	216	252	288
Blackwell GPUs (four per tray)	72	144	216	288	360	432	504	576
ConnectX-8 SuperNIC, Compute fabric (four per tray)	72	144	216	288	360	432	504	576
BlueField-3 B3240 DPU, In-band management, Customer & Storage fabric and Support Servers (one per tray)	18	36	54	72	90	108	126	144
SN5600 Ethernet switch, compute core fabric	12	12	24	24	32	32	44	44
SN5600 Ethernet switch, converged core fabric	2	2	5	6	9	10	11	11
SN2201 leaf switches for OOB management fabric.	2	4	6	8	10	12	14	16
Twin-port OSFP, 2x400 Gbps Multimode SR8, 50m Used for switch-to-switch connections (N/S Network)	20	30	192	256	320	384	448	512
MPO/APC to MPO/APC Multi-mode fibers. Switch-to-switch connections (N/S network). Length encoded by xx.	20	30	192	256	320	384	448	512
Twin-port OSFP, 2x400 Gbps Multimode SR8, 50m Used for switch-to-switch connections (E/W Network)	288	288	576	576	864	864	1152	1152
MPO/APC to MPO/APC Multi-mode fibers. Switch-to-switch connections (E/W network). Length encoded by xx.	288	288	576	576	864	864	1152	1152
100G optical modules for OOB leaf switches, transceiver, 100GbE, QSFP28, MPO, 1301nm, DR1, up to 100m	4	8	12	16	20	24	28	32
Twin-port 2x400G optical transceivers for spine switches, 2xDR4, single-mode, OSFP, 2x MPO/APC optical connectors	2	2	2	2	4	4	4	4
MPO/APC to LC Single-mode fibers. OOB MgmtSW-to-switch. One to Four breakout. Length encoded by xx.	2	2	4	4	6	6	8	8

Table 12 Endpoint connects with required transceivers, and cables

Component	Server Nodes							
	1	2	3	4	5	6	7	8
GB300 NVL72 Racks	1	2	3	4	5	6	7	8
GB300 NVL72 Compute Trays	18	36	54	72	90	108	126	144
Grace Processors (two per tray)	36	72	108	144	180	216	252	288
Blackwell GPUs (four per tray)	72	144	216	288	360	432	504	576
ConnectX-8 SuperNIC, Compute fabric (four per tray)	72	144	216	288	360	432	504	576
BlueField-3 B3240 DPU, In-band management, Customer & Storage fabric and Support Servers (one per tray)	18	36	54	72	90	108	126	144
Twin-port OSFP, 2x 400 Gbps Multimode SR8, 50m Used in switch ports for Compute Node SuperNIC-to-switch connections	72	144	216	288	360	432	504	576
Twin-port OSFP, 2x 400 Gbps Multimode SR8, 50m Used in switch ports for Compute Node DPU-to-switch connections	18	36	54	72	90	108	126	144
Twin-port OSFP, 2x 400 Gbps Multimode SR8, 50m, used in Compute Node SuperNIC for server-to-switch connections	72	144	216	288	360	432	504	576
Single-port 400 Gbps optical transceivers, multi-mode, QSFP112, used in Compute Node DPU ports for server-to-switch connections	36	72	108	144	180	216	252	288
MPO/APC to MPO/APC Multi-mode fibers. ConnectX-8 SuperNIC-to-switch. Length encoded by xx.	144	288	432	576	720	864	1008	1152
MPO/APC to MPO/APC Multi-mode fibers. DPU B3240-to-switch. Length encoded by xx.	36	72	108	144	180	216	252	288
Twin-port 2x400G optical transceivers for spine switches, 2xDR4, single-mode Used in switch ports for storage-to-switch connections	2	4	4	6	6	8	8	10
100G optical transceiver, 100GbE, QSFP28, MPO, 1301nm, DR1, up to 100m Used in upstream storage for 100Gb QSFP28 NRZ connectivity	10	18	28	36	46	54	64	72
MPO/APC to LC Single-mode fibers. Storage-to-switch. One to Four breakout. Length encoded by xx.	4	6	8	10	12	14	16	18
Twin-port 2x400G optical transceivers for spine switches, 2xDR4, single-mode Used in switch ports for customer network-to-switch connections	4	6	8	10	12	14	16	18
100G optical transceiver, 100GbE, QSFP28, MPO, 1301nm, DR1, up to 100m Used in upstream customer network for 100Gb QSFP28 NRZ connectivity	18	36	54	72	90	108	126	144
MPO/APC to LC Single-mode fibers. Customer network-to-switch. One to Four breakout. Length encoded by xx.	6	10	14	18	24	28	32	36
Twin-port OSFP, 2x 400 Gbps Multimode SR8, 50m Used in switch ports for support servers DPU-to-switch connections	12	12	12	12	12	12	12	12
Single-port 400 Gbps optical transceivers, multi-mode, QSFP112, used in support server DPU ports for server-to-switch connections	48	48	48	48	48	48	48	48
MPO/APC to 2xMPO/APC breakout multi-mode fibers. Support server ConnectX-8 400Gb (single port) to switch. Length encoded by xx.	24	24	24	24	24	24	24	24
CAT6 RJ45 cable for 1G OOB fabric connections	71	142	213	284	355	426	497	568

For exact BOM and part numbers, please work with your OEM/ODM partners and our NVIDIA support team depending on your specific rack and data center conditions.